Exp 7: Integrating a Zoho Creator Application with Another Zoho Service Using Connections and APIs

1. Aim

To integrate the gym application built in Zoho Creator with Zoho CRM by creating an authenticated connection in the Creator builder, and to write a Deluge script in the form workflow that uses the Zoho CRM API (v8) to create a Lead in CRM automatically whenever a new gym member form is submitted, then testing the integration with sample submissions.

2. Procedure

2.1 Integration Requirements

Without integration, the front desk has to copy every new gym member's details from Creator into the CRM by hand, which is slow and error-prone. The hand-off was automated with one connection and one Deluge script, as listed in Table 1.

Requirement	Component	Details
Authorise Creator to access Zoho CRM	Connection "Zoho CRM Connection"	Service: Zoho CRM. Link name: zoho_crm_connection. Created under Microservices > Connections
Run logic after a member is saved	Form workflow, Successful form submission	Form: Gym Member Form. Record event: Created. Form event: Successful form submission
Send the captured values to CRM	Deluge Map and zoho.crm.v8.createRecord	Creates one record in the Leads module using the connection
See the request and the CRM reply while testing	Deluge info statements	"CRM ACTION STARTED", "CRM RESPONSE:" and the response map

Table 1: Integration requirements and the components used

2.2 Creating the Connection to Zoho CRM

A connection lets Creator call another service's API on the user's behalf without storing a password or token in the script. It was created from the Creator builder as follows (Figure 1):

- **Open:** *Microservices* > *Connections* tab, which lists all connections available to the application.
- **Create the connection:** choose the *Zoho CRM* service, name it *Zoho CRM Connection*, and note the link name *zoho_crm_connection*. The scripts refer to the connection only by this link name.
- **Authorise:** sign in to Zoho CRM and grant the access requested, which is limited to the CRM modules the script needs to write to.
- **Check the status:** the connection card shows the type *System* and the status *Connected*. The Zoho CRM service entry beside it shows the same status.

Using a connection keeps the credentials outside the code, and the same connection can be reused by any other workflow in the application.

2.3 Mapping the Form Fields to CRM Lead Fields

Each value captured by the Gym Member Form was matched to a field of the CRM Leads module, as shown in Table 2. Last Name is a mandatory field of a CRM lead, so the customer's name was mapped to it.

Creator form field	CRM Leads field (API name)	Purpose
Customer_Name	Last_Name	Name of the lead (mandatory in CRM)
Email_Address	Email	Contact email address

Creator form field	CRM Leads field (API name)	Purpose
Contact_Number	Phone	Contact phone number
Preferred_slot	Description	Slot requested by the customer, kept as a note

Table 2: Mapping of Creator form fields to Zoho CRM Leads fields

2.4 Deluge Script: Creating the Lead in Zoho CRM

The workflow was created on the Gym Member Form with the record event *Created* and the form event *Successful form submission*, so that the CRM call is made only after the record has been saved (Figure 2). Inside the workflow, a Deluge script action holds the code below. A **Map** named leadInfo collects the field values, and **zoho.crm.v8.createRecord** sends them to the Leads module through the connection. The two **info** statements before and after the call write to the execution log, so the request and the reply can be checked while testing.

```
info "CRM ACTION STARTED";
leadInfo = Map();
leadInfo.put("Last_Name",input.Customer_Name);
leadInfo.put("Email",input.Email_Address);
leadInfo.put("Phone",input.Contact_Number);
leadInfo.put("Description",input.Preferred_slot);
response = zoho.crm.v8.createRecord("Leads",leadInfo,Map(),"zoho_crm_connection");
info "CRM RESPONSE:";
info response;
```

Script 1: Deluge script that creates a Lead in Zoho CRM using the connection

The task takes four arguments: the CRM module name (*"Leads"*), the map of field values, a map of extra parameters (left empty), and the link name of the connection (*"zoho_crm_connection"*). It returns a map, which is stored in response and printed. A successful call returns a status of success with the ID of the new record, whereas a failed call returns an error code and message, such as a missing mandatory field or insufficient connection scope.

2.5 Testing and Debugging

The integration was tested by submitting sample members through the live application, reading the log, and then opening the Leads module in Zoho CRM (Figure 3). Table 3 lists the cases covered and the behaviour the script is designed to give.

Test case	Input	Expected behaviour
Valid submission	All fields filled in the Gym Member Form	Log shows “CRM ACTION STARTED”, then “CRM RESPONSE:” with a success status and the new record ID
Lead created in CRM	Same submission	A new lead with the entered name, email, phone and preferred slot appears in the Leads module
Missing mandatory field	Customer_Name left blank in the map	CRM rejects the request and the response shows an error, which is visible in the log
Connection problem	Connection disconnected or link name misspelt	The response shows an error and no lead is created, so the connection is checked first

Table 3: Test cases for the Creator to Zoho CRM integration

3. Output

The following figures show the connection, the Deluge script and the Zoho CRM Leads module that receives the records.

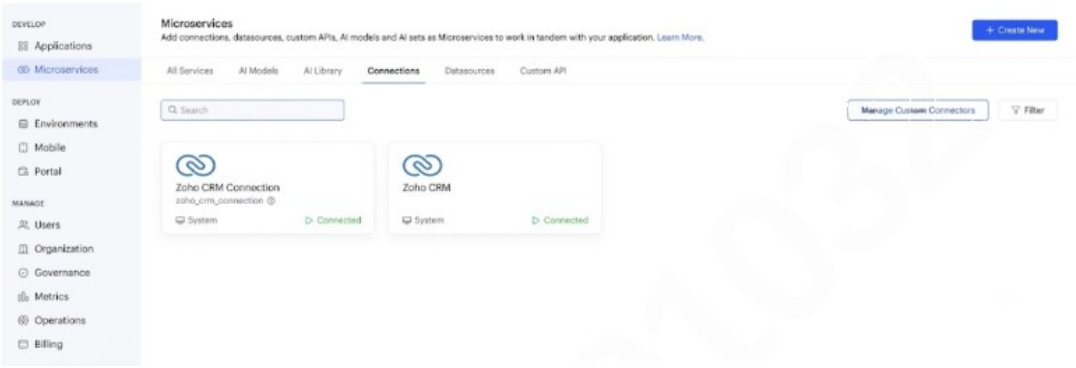


Figure 1: Microservices > Connections page showing "Zoho CRM Connection" (zoho_crm_connection) with status Connected

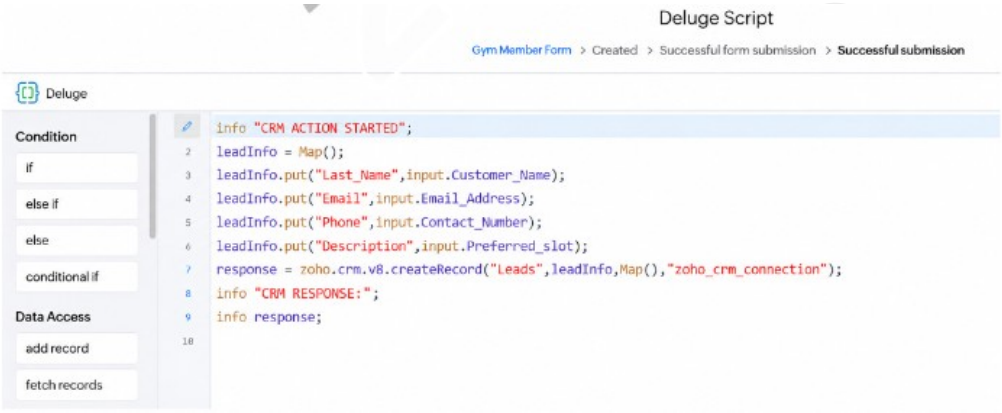


Figure 2: Deluge script on the Gym Member Form (Created > Successful form submission) that creates a Lead in Zoho CRM

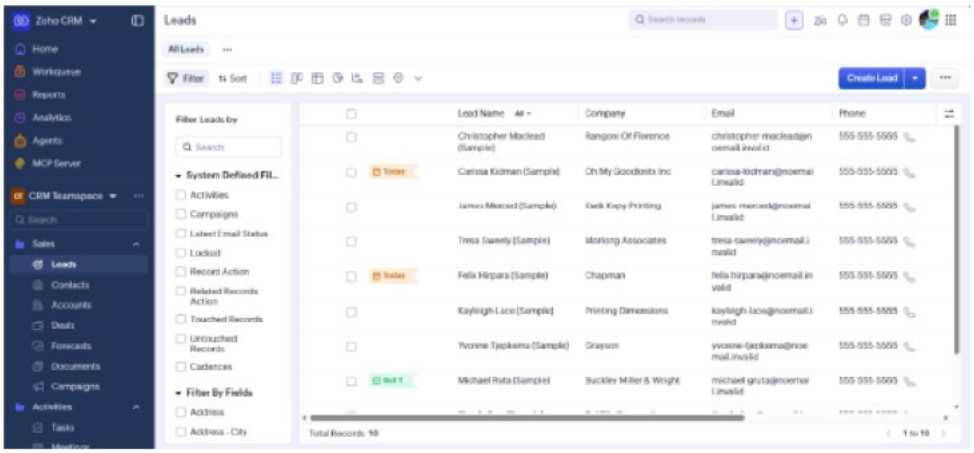


Figure 3: Leads module in Zoho CRM, the destination of the records pushed from Creator

Observations from the output

- The connection shows the status Connected, so Creator is authorised to call the Zoho CRM API without any credentials in the script.
- The workflow runs on the Created event after successful submission, so a lead is sent to CRM only for a record that was actually saved.
- The member's details are entered once in Creator and reach CRM automatically, which removes manual re-entry and the typing errors that come with it.
- The info statements around the createRecord call make the CRM reply visible in the log, which is the quickest way to trace a rejected request back to the field or connection that caused it.

4. Result

A connection to Zoho CRM was created in Zoho Creator, and a Deluge script in the Gym Member Form workflow used it with the Zoho CRM API to create a Lead in the CRM Leads module on every successful submission. The request and the CRM response were displayed with info statements for debugging. This completes the integration requirements of this experiment.

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